

HealthConnect Clinic Experience Lab

Exploratory Analysis, KPI Development & Business Insights

Week 5 Project Summary

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Data Analytics Track

This summary complements the Week 5 analytical notebook and business report. It records the planned work, completed activities, decisions, findings, challenges and proposed Week 6 direction for the HealthConnect Data Analytics track.

What I planned to accomplish

Week 5 was planned as the transition from the Week 4 analytical framework into practical implementation. The intended work was to prepare the appointment data, calculate selected no-show KPIs, carry out exploratory analysis, identify meaningful attendance patterns, develop business recommendations, and prepare outputs for dashboard development.

What I completed

- Reviewed 5,000 appointment records across 18 variables and confirmed 1,696 unique patients.
- Completed data-quality checks covering missing values, duplicates, categorical values, dates, booking lead days, appointment history and age groups.
- Created a cleaned analytical dataset and relabelled structurally blank reminder channels as No Reminder.
- Defined the main no-show analytical population as Attended + No-Show, while keeping cancellations for separate operational analysis.
- Calculated five core KPIs covering overall no-show rate, previous no-show history, reminder status, booking lead time and distance to clinic.
- Extended the EDA to appointment day, appointment time, appointment type, age group, waiting time and cancellations.
- Prepared visualisations and exported supporting KPI tables for dashboard development.
- Translated the findings into operational recommendations and documented limitations and Week 6 priorities.

3. Key findings and development outcomes

51.2% Overall no-show rate	57.8% Prior no-show history	49.9% With reminder	63.9% 31-60 day lead time
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- The overall no-show rate was 51.2% among eligible attended/no-show appointments.
- Patients with previous no-shows recorded a 57.8% no-show rate versus 46.3% for those without previous no-shows.
- Appointments without reminders recorded a 54.6% no-show rate compared with 49.9% where a reminder was sent.
- Booking lead time showed the clearest ordered pattern, increasing from 29.5% for 0-7 days to 63.9% for 31-60 days.
- The 20+ km distance group had the highest distance-based no-show rate at 60.5%.
- Follow-up appointments had the highest appointment-type no-show rate at 54.2%.
- Appointment time differences were smaller, with Evening highest at 52.6%. Waiting-time differences were generally modest apart from an unstable 60+ minute group containing only three eligible records.
- The overall cancellation rate was 5.26% and was analysed separately from no-shows.

Major challenges encountered

- Distinguishing structurally expected missing reminder-channel values from genuine data-quality gaps.
- Handling missing distance and waiting-time values without discarding otherwise useful appointment records.
- Defining an appropriate no-show denominator so that cancelled appointments were not treated as failed attendance.
- Avoiding over-interpretation of small subgroups, particularly the 60+ minute waiting-time category.
- Maintaining descriptive language because the analysis identifies associations rather than causal effects.

Important decisions and why

Decision	Reason
Cancellations excluded from no-show denominator	A cancelled appointment is operationally different from a patient failing to attend.
Blank reminder channels relabelled as No Reminder	All 1,366 blanks corresponded to reminder_sent = No, indicating structural missingness rather than incomplete data.
Distance/waiting records retained in master dataset	A missing value in one field does not make the rest of an appointment record unusable.
No-show rates used for group comparisons	Rates allow fairer comparison when groups have different appointment volumes.

Decision	Reason
Lead time and distance grouped into bands	Grouped measures are easier to interpret operationally and reveal ordered patterns.

Changes to the Week 4 approach

There was no major change to the core Week 4 problem or analytical direction. Week 5 operationalised the proposed approach by calculating the selected KPIs, expanding the EDA and clarifying specific treatment rules for cancellations, structured reminder-channel blanks and limited missing distance/waiting-time values. The analysis also showed that previous no-show history, booking lead time, and distance were more informative than some contextual variables such as appointment time and waiting time.

Remaining work

- Complete and refine the Power BI dashboard using the exported KPI tables and cleaned dataset.
- Add genuine cross-track collaboration evidence if completed.
- Check whether the strongest patterns persist when variables are combined or segmented further.
- Review dashboard narrative, labels and KPI definitions for consistency with the notebook.
- Prepare the project for Week 6 refinement and subsequent final presentation stages.

Proposed focus for Week 6

- Test interactions such as booking lead time by reminder status and appointment type.
- Refine and validate the strongest Week 5 findings across subgroups.
- Improve the Power BI dashboard and business storytelling.
- Validate KPI definitions and calculation logic.
- Incorporate documented cross-track feedback.
- Evaluate whether the Week 5 recommendations remain supported across the relevant segments.
- Develop a clearer final analytical narrative for the next project stage.

Closing Summary

Week 5 successfully moved the HealthConnect project from planning to implementation. The appointment dataset was cleaned and validated, five core KPIs were calculated, exploratory analysis was completed, and the strongest no-show patterns were converted into practical hypotheses for operational testing. The analysis now provides a clear foundation for dashboard refinement, deeper segmentation, and cross-track collaboration in Week 6.